

**Greener Data Centers: How Custom Chips Can Power Sustainable AI and Blockchain
Systems**

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Abstract

The rapid growth of artificial intelligence and blockchain technologies has led to a surge in data center energy consumption, raising concerns about their environmental impact. This literature review presentation examines the use of application-specific integrated circuits (ASICs) - computer chips designed for specialized tasks - can help the carbon footprint associated with these AI and blockchain systems. ASICs, which are custom designed for specific tasks, have been explored as an alternative to general-purpose processors such as GPUs and CPUs due to their potential for higher efficiency and lower power use. Studies across engineering and blockchain computing provide arguments to how ASICs tailored for AI or blockchain can reduce redundant computation and improve overall energy efficiency. However, challenges remain around hardware flexibility, manufacturing costs, overall sustainability, and environmental impact. Ultimately, this work aims to assess the potential of ASIC technology to make AI and blockchain infrastructures more environmentally responsible and to highlight directions for future research.

Keywords: ASIC design; energy-efficient computing; artificial intelligence hardware; blockchain systems; sustainable data centers; low-power architecture; hardware acceleration; carbon footprint reduction

Introduction

Artificial Intelligence (AI) and blockchain systems have become preeminent technologies in modern digital infrastructure. They power applications ranging from autonomous vehicles to decentralized finance. But the computational demands for both systems have increased global energy consumption, raising serious concerns about their environmental impact. As AI models like ChatGPT grow more complex and blockchain networks scale, the data centers that support them have become major consumers of electricity, mainly due to the heavy reliance on graphics processing units (GPUs) and central processing units (CPUs). These processors are often energy-inefficient for specialized cases, performing redundant computations that contribute to higher costs and carbon emissions.

This growing awareness has driven research towards a new technology, application-specific integrated circuits (ASICs). These are custom computer chips designed for specific computational tasks. Unlike CPUs and GPUs, ASICs can execute specific operations with high efficiency and less energy draw. Widespread adoptions of these can potentially dramatically reduce power usage and heat dissipation from data centers. In both AI and blockchain contexts, ASICs can improve efficiency and accuracy by tailoring the hardware to be in line with calculations related to machine learning and cryptography – both of which are heavily sources of calculation for AI and blockchain applications.

The purpose of this literature review is to assess how ASICs can reduce the carbon footprint associated with AI and blockchain systems. It explores the trade-offs between flexibility and efficiency, while highlighting recent advances in ASIC-based architectures, and identifies open challenges in sustainability and scalability. Furthermore, it connects research findings from

energy-efficient computing, hardware acceleration, and embedded AI to demonstrate how ASIC adoption can reshape the environmental cost of data centers worldwide.

Methodology

The types of sources included in this literature review are peer-reviewed scholarly publications, research focusing on energy-efficient or application-specific computing architectures, studies examining hardware acceleration in AI or blockchain systems, and articles that discuss sustainability or carbon emissions related to computational hardware.

All of these sources were found using Purdue Library using key terms that can be found in the abstract for this literature review. Many of these studies are from databases such as ProQuest, IEEE Xplore, and ACM Digital Library.

Energy Demands of AI and Blockchain

The exponential growth of AI computation, especially in large-language models (LLMs) such as ChatGPT, has strained existing infrastructure. GPUs, though capable of processing immense amounts of data at multiple times, are not optimized for every operation involved in training large-language models. Similarly, blockchain operations related to obtaining cryptocurrencies like Bitcoin rely on massive clusters of GPUs working in tandem to perform repetitive mathematical proofs and hashing functions. These all consume immense amounts of energy that take a toll on the electricity grid. According to Liu and Zhai (2025), the integration of AI computation in high-performance computing (HPC) systems can lead to immense carbon emissions, particularly when power sources are not renewable. This AI and blockchain framework demonstrate that without changes made to the hardware, the carbon footprint of AI and blockchain will continue to escalate, even as renewable forms of energy are used.

Blockchain, particularly obtaining Bitcoin – which is called mining - mirrors this trend. Mechanisms used to make mining Bitcoin safe depend on solving cryptographic puzzles. This requires hardware to perform trillions of hash calculations per second. The initial use of CPUs and GPUs for mining quickly gave way to specialized ASIC miners due to their superior efficiency and speed. However, these changes concentrated computational power in the hands of large mining operations, raising environmental and economic concerns about sustainability and decentralization.

GPU Limitations and Energy Inefficiency

GPUs were originally made for rendering graphics, and not necessarily for optimizing energy performance. Their general-purpose architecture allows flexibility across different AI

models but results in redundant operations and high-power consumption. Qiu et al. (2022) emphasize that despite improvements in GPU energy scaling, AI workloads are inherently power-intensive due to the large matrix operations and memory access patterns they require.

In data centers, this inefficiency compounds across thousands of machines. Cooling systems, often responsible for 30-40% of total energy use, must dissipate heat generated by GPU clusters. As Sojonen and Pietri (2025) discuss, the design of efficient computing systems for AI and autonomy increasingly depends on hardware accelerators that minimize unnecessary energy usage. Their analysis notes that a shift toward efficient hardware design, could substantially reduce this inefficiency, aligning performance with environmental goals.

Thus, while GPUs remain essential for flexible computation, their lack of specialization poses a fundamental barrier to sustainable scaling. This challenge has spurred the exploration of ASICs, which trade flexibility for efficiency.

ASIC Design and Efficiency Improvements

ASICs are computer chips built to execute specific functions with optimized logic, minimal transistor count, and reduced energy with transistor switching. By eliminating general-purpose control logic and cache hierarchies, ASICs achieve performance advantages that surpass GPUs by several orders of magnitude. Kim, Lee, and Choi (2024) demonstrate this efficiency in embedded AI systems, where proactive boost mechanisms dynamically adjust computing performance according to workload requirements. Their computing approach, which normalizes performance to power usage, shows that ASIC-based systems can sustain high rates with low thermal output, making them ideal for portable or distributed AI applications.

In AI datacenters, similar principles apply. ASIC accelerators such as Google's Tensor Processing Units (TPUs) and other AI-specific chips show how dedicated matrix multiplication units can reduce power draw. These chips often incorporate memory on-chip to minimize energy usage when transferring data, which GPUs lack. ASICs can therefore directly contribute to reducing the energy footprint of LLM training.

Sachdeva et al. (2016) provides validation of ASIC efficiency in the design of low-power very large-scale integration (VLSI) systems for fuzzy logic control. These results highlight how custom hardware can outperform general-purpose alternatives by eliminating repetitive calculations, supporting the case for ASICs in energy-sensitive contexts.

Applications in AI and Blockchain

ASICs have proven their potential most visibly in blockchain mining. As Bitcoin's network difficulty increased, GPU-based miners could no longer compete with ASIC-based hardware. These ASIC miners achieve exponential improvements in their calculation rate per unit of power (watt). This improvement drastically reduced the energy need for each cryptographic puzzle computation. Yet, their overall network consumption remains high because miners continually increase calculation power to remain profitable with their Bitcoin. This dynamic illustrates the promise and tradeoff with ASIC efficiency. The cost of energy decreases per unit that uses ASIC, but the total system's consumption may remain high if scaling of ASIC units increase.

In AI, ASICs have increasingly been deployed in edge computing and TinyML (Tiny Machine Learning) environments where energy budgets are nonconsequential. The article *Advancements in TinyML: Applications, Limitations, and Impact on IoT Devices* (2023) notes

that embedded ASIC designs for machine learning enable low-power sensors on devices such as sensors and wearable technology. By processing data locally rather than in the cloud, these systems minimize energy from data transmission and reduce the need for power-hungry datacenter computations. Xu et al. (2022) similarly verified the design of an AI vision chip for edge computing, showing that ASIC based processing can achieve high efficiency through hardware simulation.

These studies highlight ASIC's versatility. Whether supporting edge AI or securing blockchain transactions, the technology's defining advantage is its ability to run specific computations for specific use cases.

Sustainability Challenges and Trade-Offs

Despite these advantages, ASICs introduce new sustainability concerns. Manufacturing ASICs require significant upfront investment and specialized fabrication facilities, which can produce high carbon emissions. Once produced, ASICs cannot be reprogrammed or repurposed. This would mean old designs can contribute to waste. Kim et al. (2024) and Liu and Zhai (2025) both acknowledge that sustainable computing cannot rely solely on achieving efficiency metrics; it must consider impact over multiple years, recyclability, and integration with renewable energy sources.

Moreover, the ASIC model introduces issues of accessibility and centralization. In blockchain contexts, ASIC mining has concentrated computational power within a small number of operators. This contracts the decentralized philosophy of blockchain technology. Qiu et al. (2022) warns that energy-efficient design must account for security and economic implications.

At the AI infrastructure level, integrating ASICs into data centers demands redesigns. Sojonen and Pietri (2025) argue that the path towards efficiency involves both hardware redesigns and software redesigns. They suggest that new algorithms can align with ASIC design principles to further reduce redundant calculations. Thus, while ASICs can mitigate immediate energy inefficiencies, achieving long-term sustainability required a solution across hardware, software, and policy.

Conclusion

This literature review shows that ASICs can hold significant promise for reducing the environmental impact of AI and blockchain technologies. By tailoring hardware to specific computational workloads, ASICs can outperform GPUs and CPUs in both speed and energy efficiency, directly addressing the growing carbon footprint of data centers. Studies used in this review demonstrate that combining ASICs with improved management of power can yield gains in sustainability. Practical applications illustrate that ASIC efficiency is possible and useful.

However, the transition to ASIC-based infrastructures introduces challenges that cannot be overlooked. Limited flexibility, high manufacturing costs, and potential electronic waste from rapid changes in ASICs can raise new concerns. Blockchain's reliance on ASICs has concentrated computing resources, which proves as a warning for ASIC-based infrastructure for LLM training.

Future research should pursue holistic studies that combine ASIC design with renewable energy integration, recyclability, and optimizations of software. Investigating these new processes could bridge the gap between performance and environmental responsibility. As AI and blockchain continue to define the technology landscape, ASIC technology – guided by sustainable design – could become the future of a green computation era.

References

Advancements in TinyML: Applications, Limitations, and Impact on IoT Devices. (2023).

Proquest.com. <https://www.proquest.com/docview/3103840486?accountid=13360&pq-origsite=primo&sourcetype=Scholarly%20Journals>

This article explores the recent developments in TinyML and how it enables machine learning on resource-constrained IoT devices. It discusses the trade-offs between power efficiency and computational capability while identifying bottlenecks. The paper emphasizes TinyML's role in IoT systems becoming more autonomous and energy efficient.

Feng, T.-T., Li, Y., Zhong, C., Wang, H.-Q., & Zhang, N.-Q. (2025). How Does Electricity Consumption in Blockchain Applications Impact Resource Consumption and Environmental Emissions? *International Journal of Energy Research*, 2025(1).

<https://doi.org/10.1155/er/5772901>

This article explores the environmental consequences of blockchain systems by modeling their electricity usage and related carbon emissions. The study quantifies how resource-intensive blockchain mining and validation processes contribute to global energy demand. Their findings highlight the urgent need for renewable energy adoption and efficiency improvements in blockchain.

Kim, J., Lee, G., & Choi, H. (2024). Energy-Efficient Heterogeneous Computing via Normalized Performance based Proactive Boost for Embedded Artificial Intelligence. *2023 IEEE International Conference on Consumer Electronics (ICCE)*, 1–6.

<https://doi.org/10.1109/icce59016.2024.10444347>

This conference paper presents a new mechanism to enhance energy efficiency in embedded AI systems. By balancing workload distribution through normalized metrics,

the new mechanism achieves faster computational speed at a reduced power cost. This approach contributes to advancing TinyML and embedded AI for low-power applications.

Liu, H., & Zhai, J. (2025). Carbon Emission Modeling for High-Performance Computing-Based AI in New Power Systems with Large-Scale Renewable Energy Integration. *Processes*, 13(2), 595–595. <https://doi.org/10.3390/pr13020595>

Liu and Zhai develop a carbon emissions model for AI systems operating on high-performance computing clusters. Their research integrates renewable energy data to assess how green power sources affect overall emissions levels. The work provides valuable insights for designing more sustainable infrastructure.

Meemken, B., & Poth, A. (2024). Eco-Friendly AI: A Guide to Energy-Efficient Techniques Across the AI Life-Cycle. *Communications in Computer and Information Science*, 36–50. https://doi.org/10.1007/978-3-031-71142-8_3

Meemken and Poth discuss an overview of energy-efficient methods across the AI life cycle, from algorithm design to deployment. The chapter discusses hardware optimization and quantization techniques that balance performance with environmental responsibility. It offers a holistic framework for designing AI systems that minimize energy use without sacrificing accuracy.

Mikko Sojonen, & Pietri, S. (2025). Lecture Review in Central Texas—Efficient Computing for Artificial Intelligence and Autonomy: From Hardware Accelerators to Algorithm Design [Chapters]. *IEEE Solid-State Circuits Magazine*, 17(1), 158–159. <https://doi.org/10.1109/mssc.2024.3497532>

Sojonen and Pietri summarize recent advancements in AI computing, focusing on

hardware accelerators and software optimization. The review discusses the synergy between semiconductor innovation and AI computing performance.

Patrizio, A. (2022, April 8). *Intel claims sustainability for its custom chip that mines bitcoins faster than GPUs: Intel's Blockscales ASIC designed specifically for blockchain hash operations with greater power efficiency is available this fall*. Proquest.com.

<https://www.proquest.com/trade-journals/intel-claims-sustainability-custom-chip-that/docview/2648615438/se-2?accountid=13360>

Patrizio reports on Intel's Blockscales ASIC chip, designed to perform blockchain computations with increased energy efficiency compared to GPUs. The article emphasizes the company's focus on sustainability and reduced power draw in cryptocurrency mining.

Qiu, M., Xu, K., Zhang, C., & Zhang, T. (2022). Introduction to the Special Section on Energy-efficient and Secure Computing for Artificial Intelligence and Beyond. *ACM Transactions on Sensor Networks*, 18(4), 1–3. <https://doi.org/10.1145/3558553>

Qiu and colleagues introduce a special section exploring the intersection of security and energy efficiency in AI computing. The authors highlight emerging research focused on low-power architectures and secure edge deployment. Their introduction establishes a framework for understanding how energy efficiency is becoming central to trustworthy AI systems.

Sachdeva, E., Pratik Porwal, Nalini Vidyulatha, & Shrestha, R. (2016). *Design of low power VLSI-architecture and ASIC implementation of fuzzy logic based automatic car-parking system*. 12, 1–6. <https://doi.org/10.1109/indicon.2016.7839149>

This study presents a low-power VLSI and ASIC-based design for an intelligent car-

parking system using fuzzy logic. The authors demonstrate how custom hardware can minimize power consumption while maintaining decision accuracy.

XU, X., NING, K., ZHENG, X., ZHAO, M., XU, M., WU, N., & LIU, L. (2022). Verification of an artificial intelligence vision chip design for edge computing based on hardware simulation system. *物联网学报*, 6, 20–28.

<https://doaj.org/article/6eed847a952b4f70beb22a04c39718c8>

Xu et al. look into an AI vision chip architecture optimized for edge computing through hardware simulation. Their results demonstrate high performance at significantly reduced energy costs, supporting efficient real-time image processing.